Scaling Hierarchical Coordination for Million-Unit Space Swarms: Discrete Event Simulation and Architectural Analysis

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***Abstract*—Emerging mega-constellations such as Starlink, with over 7,000 operational satellites and 12,000 approved as of 2024, represent the current frontier of coordinated space systems. Proposed large-scale space infrastructure, including solar power satellite arrays, will require coordination of 10^5 to 10^6 autonomous nodes—scales at which no existing coordination architecture has been systematically evaluated. We present a discrete event simulation (DES) study comparing centralized, hierarchical, and mesh coordination topologies at scales from 10^3 to 10^6 autonomous spacecraft. Our results demonstrate that hierarchical coordination is the only viable topology at million-node scale: centralized architectures, modeled as a worst-case single-server bound, encounter processing latency bottlenecks at approximately 10,000 nodes (a limit that shifts upward with parallelization), while mesh topologies—when parameterized for the global state convergence required by fleet-wide collision avoidance—incur $O(N^2)$ gossip overhead exceeding 25% of available bandwidth beyond 100,000 nodes. The hierarchical approach, employing clusters of 50–100 nodes with rotating coordinators, scales past 10^6 nodes with communication overhead of only 2–8%. We identify the optimal coordinator duty cycle as 24–48 hours, balancing handoff overhead against single-point-of-failure exposure, and observe a parameter-dependent overhead inflection near 50,000 nodes beyond which advanced optimizations become necessary. An AI-assisted architectural exploration using three large language models generated the Shepherd/Flock heterogeneous hardware concept and other design proposals that are consistent with the simulation findings. We discuss applicability to mega-constellation management, autonomous drone swarms, and distributed sensor networks.**

***Index Terms*—Space systems coordination, swarm robotics, hierarchical control, discrete event simulation, mega-constellations, autonomous spacecraft, multi-agent systems**

I. Introduction

A. The Coordination Scaling Problem

The coordination of large numbers of autonomous spacecraft represents one of the foremost unsolved challenges in aerospace systems engineering. The largest operational constellation today, SpaceX's Starlink network, manages approximately 7,000 active satellites (as of mid-2024) through centralized ground-based coordination [1]. While this approach has proven effective at current scale, the system already requires substantial ground infrastructure and has encountered coordination challenges during conjunction events [2]. Approved expansion plans for Starlink alone envision 42,000 satellites, and other operators including Amazon's Project Kuiper and OneWeb are deploying their own constellations [3], [4].

Looking further ahead, proposed space infrastructure concepts—including space-based solar power arrays and orbital manufacturing facilities—contemplate fleets of 10^5 to 10^6 autonomous units in the near term, with aspirational concepts such as Dyson swarm precursors potentially requiring far larger fleets in the long term [5], [6]. At these scales, the coordination problem is not merely a larger version of current constellation management; it is qualitatively different. Message processing latency, bandwidth allocation, state synchronization, and failure recovery all exhibit nonlinear scaling behaviors that can render architectures adequate at 10^3 nodes entirely unworkable at 10^6 [7].

No prior work has systematically compared coordination architectures across this range of scales using quantitative simulation. The literature on swarm robotics has extensively studied coordination at scales of tens to hundreds of agents [8], and constellation management literature addresses systems of up to approximately 10,000 nodes [9], but the intermediate regime of 10^4 to 10^6 nodes remains largely unexplored.

B. Research Questions

This study addresses three interrelated research questions:

RQ1: How do centralized, hierarchical, and mesh coordination architectures scale in terms of communication overhead, message latency, and failure resilience as fleet size increases from 10^3 to 10^6 nodes?

RQ2: What is the optimal duty cycle for rotating cluster coordinators in a hierarchical architecture, considering the trade-off between handoff overhead and single-point-of-failure exposure?

RQ3: At what fleet size do coordination constraints begin to dominate operational overhead, and what optimizations are necessary beyond this threshold?

C. Contributions

The principal contributions of this paper are:

- The first comparative discrete event simulation of coordination topologies at scales spanning three orders of magnitude (10^3 to 10^6);
- Quantified communication overhead, message processing latency, and failure resilience characteristics for each topology as a function of scale;

- Identification of an optimal 24–48 hour coordinator duty cycle with associated power and reliability trade-off characterization;
- Establishment of a 50,000-node inflection point beyond which advanced optimizations become necessary;
- AI-assisted architectural exploration through a multi-model deliberation process, yielding the Shepherd/Flock heterogeneous hardware concept; and
- Practical design guidelines applicable to near-term mega-constellation management.

II. Related Work

A. Constellation Coordination

Current satellite constellation management relies predominantly on centralized ground-based coordination. SpaceX operates Starlink through a network of ground stations and a centralized mission control that computes orbit adjustments, manages conjunction avoidance, and coordinates spectrum sharing [1]. The European Space Agency's Space Debris Office coordinates conjunction warnings across multiple operators, but the computational burden scales linearly with constellation size [2]. OneWeb employs a similar centralized paradigm for its constellation management [4]. All of these systems operate at fewer than 15,000 nodes, and none has published evidence of architectural scalability beyond this regime.

The Iridium constellation, while smaller at 66 active satellites, introduced inter-satellite links (ISLs) that reduced dependence on ground control for routine operations [22]. This partial decentralization foreshadowed the architectural direction explored in the present study, although Iridium's crosslink topology is fixed and would not scale to large swarm configurations.

B. Swarm Robotics

The swarm robotics literature provides a rich foundation for decentralized coordination. Brambilla et al. [8] present a comprehensive taxonomy of swarm behaviors, noting that most experimental demonstrations involve 10–100 agents with a few reaching 1,000. Dorigo et al. [11] survey the state of the art in swarm engineering, emphasizing the gap between theoretical scalability claims and experimental validation. Reynolds' seminal work on flocking behaviors [10] demonstrated that simple local rules can produce emergent global coordination, but the computational analysis assumed homogeneous agents with identical sensing capabilities—an assumption that breaks down when heterogeneous coordination roles are required.

Bio-inspired approaches including ant colony optimization [12], particle swarm optimization [13], and artificial bee colony algorithms [14] have been applied to multi-agent coordination. However, these approaches are primarily optimization heuristics rather than operational coordination architectures, and their convergence properties at 10^6-node scale have not been established.

C. Mathematical Foundations

Several theoretical frameworks underpin the analysis of large-scale coordination. Lamport's foundational work on logical clocks and event ordering in distributed systems [32] establishes the theoretical basis for consistent coordination in networks without global time references—directly relevant to space swarms where light-speed delays preclude tight synchronization. Consensus protocols such as Raft [33] provide mechanisms for coordinator election and state replication in the presence of failures, and inform the coordinator rotation scheme explored in our hierarchical model. Tolstaya et al. [15] demonstrated that Graph Neural Network (GNN) controllers for multi-robot systems achieve communication complexity that scales linearly with the number of agents, contrasting with the quadratic scaling of naive broadcast approaches. This result provides a theoretical lower bound on the communication overhead achievable by any distributed coordination scheme.

Mean-field game theory, introduced by Lasry and Lions [16] and independently by Huang et al. [17], offers a framework for analyzing systems with very large numbers of interacting agents by replacing individual agent tracking with statistical population descriptions. This approach is particularly relevant at the scales considered in this study, where tracking the state of every individual node becomes computationally intractable.

Recent work on decentralized control with limited communication by Li et al. [18] establishes bounds on the time required for local information to propagate globally through a decentralized network, demonstrating that hierarchical topologies achieve $O(\log N)$ propagation time compared to $O(N)$ for flat mesh networks and $O(1)$ for centralized (at the cost of a central bottleneck). These theoretical results motivate the hierarchical topology explored in our simulation study.

D. Military Drone Swarm Programs

The DARPA Offensive Swarm-Enabled Tactics (OFFSET) program has demonstrated coordination of over 100 autonomous drones in field exercises [19]. The Naval Research Laboratory has conducted experiments with up to 250 coordinated UAVs [20]. The Department of Defense Replicator initiative aims to field thousands of autonomous systems [21]. While these programs represent the state of the art in operational drone swarm coordination, none has published results at scales exceeding approximately 250 units, and the coordination architectures employed are typically centralized or use fixed hierarchical command structures.

III. Simulation Framework

A. Discrete Event Simulation Architecture

We implement a discrete event simulation (DES) framework in which autonomous nodes exchange messages according to the rules of one of three coordination topologies. The simulation is event-driven: no computation occurs between events, allowing efficient modeling of long operational periods. Each simulation run models one year of swarm operations.

The event types modeled are:

- **Status reports**: periodic health and position telemetry from nodes to their coordinator (or broadcast, depending on topology);
- **Coordination messages**: task assignments, orbit adjustments, and cluster-level directives;
- **Handoff events**: transfer of coordinator responsibility between nodes (hierarchical topology only);
- **Failure events**: node failures, drawn from an exponential distribution;
- **Collision avoidance**: proximity alerts requiring time-critical response.

The simulation clock operates at one-second resolution for collision avoidance events and one-minute resolution for routine coordination, reflecting the differing time-criticality of these operations. Messages experience propagation delay proportional to inter-node distance (at light speed) plus processing delay at the receiving node, modeled as a service time drawn from the appropriate queueing model for each topology. We note that the communication model abstracts away MAC-layer scheduling, link acquisition, and pointing constraints that are significant in real space links [34]; these simplifications are discussed in Section VI-D.

B. Topology Models

1) Centralized Topology: In the centralized model, a single ground-based coordinator receives all status reports, computes coordination decisions, and issues commands to individual nodes. We model the coordinator as an $M/D/1$ queue with finite buffer [23]:

$$\rho = \frac{\lambda}{\mu} = \frac{N \cdot r}{C} \tag{1}$$

where N is the number of nodes, r is the per-node reporting rate, C is the coordinator's processing capacity (messages per second), and ρ is the server utilization. The mean waiting time in the $M/D/1$ queue is:

$$W_q = \frac{\rho}{2\mu(1-\rho)} \tag{2}$$

As $\rho \to 1$, waiting time diverges. Our simulation parameterizes $C = 1{,}000$ messages per second for a single processing server, representative of a single ground station processing thread. At a reporting rate of $r = 0.1$ messages per second per node, utilization reaches $\rho = 1.0$ at $N = 10{,}000$.

This single-server model represents a *worst-case bound* on centralized processing capacity. Real ground systems employ parallel processing across multiple servers, which can be modeled as an $M/D/c$ queue where c is the number of parallel servers. In this case, utilization becomes:

$$\rho_c = \frac{N \cdot r}{c \cdot C} \tag{3}$$

and the scalability limit shifts to $N = c \cdot C/r$. With $c = 10$ parallel servers, the centralized architecture could support approximately 100,000 nodes before saturation. Our single-server parameterization ($c = 1$) therefore represents the architectural worst case; we adopt it to establish a conservative baseline and to highlight the *qualitative* bottleneck inherent in centralized processing, which persists regardless of c since the limit remains linear in the number of servers while fleet growth may be exponential.

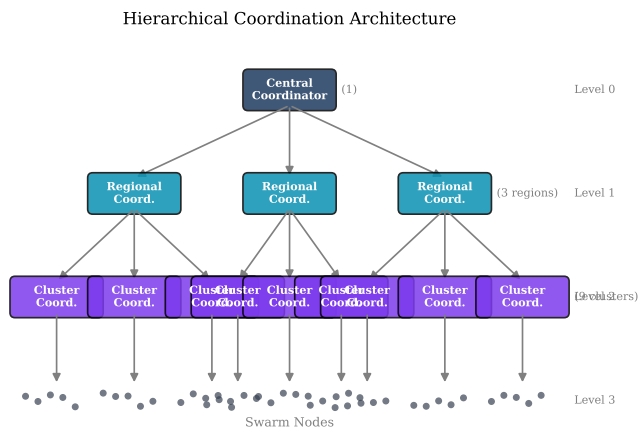

Fig. 1. Four-level hierarchical coordination architecture. Ground control connects to 10–100 regional coordinators, each managing 10–100 cluster coordinators, each responsible for 50–100 individual nodes. Labels indicate message aggregation ratios at each level of the hierarchy.

The total message complexity per coordination cycle is $O(N)$, as each node sends one report and receives one command.

2) Hierarchical Topology: The hierarchical model implements a four-level tree structure (Fig. 1):

$$\text{Levels: Ground} \to \text{Regional} \to \text{Cluster} \to \text{Node} \tag{4}$$

with configurable fan-out at each level. Nodes report to their cluster coordinator, which aggregates reports and forwards summaries to the regional coordinator, which in turn aggregates for the ground station. The aggregation ratio α at each level reduces message volume:

$$M_{\text{total}} = N + \frac{N}{k_c} + \frac{N}{k_c \cdot k_r} \tag{5}$$

where k_c is the cluster size and k_r is the number of clusters per regional coordinator. For $k_c = 100$ and $k_r = 100$, the ground station processes only $N/10{,}000$ messages per cycle rather than N. With the fixed four-level hierarchy described above, the total message complexity per coordination cycle is $O(N)$: each level contributes $O(N)$ messages, and the number of levels is constant. This linear scaling is a key strength of the fixed-depth design, ensuring that coordination cost grows proportionally with fleet size without the logarithmic overhead that would arise if hierarchy depth were adapted dynamically. An adaptive hierarchy whose depth grew as $O(\log N)$ with fleet size would yield $O(N \log N)$ total complexity; we do not employ such a design, as the fixed four levels provide sufficient aggregation across the simulated range of 10^3 to 10^6 nodes.

Coordinator rotation is modeled as a state transfer event. The outgoing coordinator transmits its accumulated state (10–50 MB, depending on cluster size) to the incoming coordinator

over an optical inter-satellite link at 1–10 Gbps, completing the handoff in 1–10 seconds. During the handoff window, the cluster operates in a degraded mode where routine coordination is deferred but collision avoidance remains active through direct node-to-node communication.

3) Mesh Topology: The mesh model implements a gossip protocol in which each node periodically selects a random subset of f neighbors (the *fanout*) and exchanges state information. In standard epidemic dissemination with constant fanout $f = O(1)$ or logarithmic fanout $f = O(\log N)$, convergence to full state dissemination requires $O(\log N)$ rounds, yielding total message complexities of $O(N \log N)$ or $O(N \log^2 N)$ respectively [24].

However, the coordination scenario considered here requires *global state convergence*: each node must maintain awareness of its neighbors' trajectories to support fleet-wide collision avoidance (a safety-critical requirement). In a swarm of N nodes distributed across an orbital shell, each node must eventually receive trajectory updates from all $O(N)$ nodes—not merely propagate a single rumor. When the state to be disseminated is the full N-node trajectory table, the total information that must flow through the network is $O(N^2)$ regardless of the gossip protocol used, because each of N nodes must receive $O(N)$ state entries. With fanout $f = O(N/\log N)$, convergence occurs in $O(\log N)$ rounds:

$$M_{\text{mesh}} = O(N \cdot f \cdot \log N) = O(N^2) \tag{6}$$

A local-only mesh protocol using constant fanout $f = k$ (where k is the number of orbital neighbors) would achieve $O(kN)$ per-round complexity and could support local coordination effectively. However, such a protocol cannot provide the global trajectory awareness needed for fleet-wide collision avoidance and coordinated orbit-raising maneuvers—operations that require information from nodes outside the local neighborhood. The $O(N^2)$ characterization thus reflects the *information-theoretic cost of global coordination in a flat topology*, not an unfavorable choice of gossip parameters. This distinction is important: the hierarchical topology avoids this cost through aggregation, reducing the per-node information requirement from $O(N)$ to $O(k_c)$ (the cluster size).

Convergence time scales with the network diameter D:

$$T_{\text{converge}} = D \cdot \tau_{\text{gossip}} \tag{7}$$

where τ_{gossip} is the gossip round interval. For a random geometric graph with N nodes in three-dimensional orbital space, $D = O(N^{1/3})$.

C. Node Model

Each simulated node is characterized by the following parameters:

- **Communication power**: 5 W baseline, 15–20 W in coordinator mode;
- **Bandwidth allocation**: 1 kbps per node for coordination traffic;
- **Failure model**: exponential inter-arrival times with a 2% annual failure rate, yielding a mean time to failure (MTTF) of 50 years per node;
- **State vector**: position (3 components), velocity (3 components), health status, power level, and cluster assignment.

The failure rate of 2% annually is consistent with observed rates for modern small satellites in low Earth orbit [25]. Upon failure, a node ceases all communication; in the hierarchical topology, the cluster coordinator detects the failure within one reporting cycle and reassigns responsibilities.

D. Monte Carlo Framework

To account for stochastic variability in failure events, message timing, and initial conditions, we employ a Monte Carlo approach with 50–100 independent runs per configuration. The swept parameter space includes:

- Node count: $N \in \{10^3, 5 \times 10^3, 10^4, 5 \times 10^4, 10^5, 5 \times 10^5, 10^6\}$;
- Cluster size (hierarchical only): $k_c \in \{50, 75, 100, 150, 200, 300, 500\}$;
- Coordinator duty cycle (hierarchical only): $\tau_d \in \{1, 6, 24, 48, 168\}$ hours;
- Gossip fanout (mesh only): $f \in \{3, 5, 10, 20, \sqrt{N}\}$.

The primary metrics collected are: communication overhead (fraction of total bandwidth consumed by coordination traffic), mean and 99th-percentile message processing latency, coordination success rate (fraction of coordination cycles completed within deadline), and system availability (fraction of time each node is reachable by the coordination system). Results are reported as means with 95% confidence intervals computed via the bootstrap method (10,000 resamples with bias-corrected and accelerated intervals).

E. Simulation Parameters Summary

Table I provides a comprehensive listing of all key simulation parameters to support reproducibility. The simulation is implemented in Python using an event-driven architecture. Wall-clock runtime ranges from approximately 2 minutes for $N = 10^3$ configurations to approximately 18 hours for $N = 10^6$ on a 64-core workstation. Each Monte Carlo configuration uses independent random seeds drawn from a Mersenne Twister generator seeded with the configuration index.

IV. Results

A. Topology Comparison

Table II and Fig. 2 summarize the scaling behavior of the three coordination topologies across the simulated range.

The centralized topology maintains acceptable overhead (5–15%) up to approximately 10,000 nodes under our single-server parameterization ($c = 1$), beyond which message processing latency at the central coordinator diverges (Fig. 3). This bottleneck arises from processing latency, not bandwidth: even with ample communication bandwidth, the sequential processing of status reports at the central coordinator introduces queueing delays that exceed coordination cycle deadlines. At $N = 10{,}000$ with $c = 1$, the $M/D/1$ queue utilization ρ approaches unity (Eq. 1), and mean waiting time

TABLE I
SIMULATION PARAMETERS

Parameter	Value	Notes
Message Model		
Status report size	256 B	Position, velocity, health
Coordination msg size	512 B	Task assignment, orbit adj.
Collision avoidance msg	128 B	Priority alert
Handoff state size	10–50 MB	Scales with k_c
Status reporting rate r	0.1 msg/s	Per node
Collision avoidance rate	10^{-4}/node/s	Poisson process
Link and Processing		
Per-node bandwidth	1 kbps	Coordination channel
Optical ISL capacity	1–10 Gbps	For handoff transfers
Coordinator capacity C	1,000 msg/s	Per server
Message buffer size	10,000 msg	Drop-tail on overflow
Failure Model		
Node failure rate	2%/year	Exponential distribution
MTTF per node	50 years	Operational phase only
Coordinator failure	Same as node	No enhanced reliability
Monte Carlo Configuration		
Runs per config	50–100	100 for $N \geq 10^5$
Simulation duration	1 year	Per run
Clock resolution	1 s / 60 s	Collision / routine
Random seed strategy	Sequential	Mersenne Twister

TABLE II
COORDINATION TOPOLOGY COMPARISON

Architecture	Scalability Limit	Overhead	Failure Mode
Centralized	∼10,000	5–15%	Single point
Hierarchical	1,000,000+	2–8%	Graceful
Mesh	∼100,000	10–25%	Excellent

(Eq. 2) exceeds the coordination cycle period. As noted in Section III-B-1, parallelization via $M/D/c$ queues shifts this limit to $N = c \cdot C/r$; however, the fundamental architectural constraint—that all coordination decisions flow through a finite set of ground servers—persists, making centralized architectures progressively more expensive to scale compared to hierarchical alternatives.

The mesh topology provides excellent failure resilience—the loss of any individual node has negligible impact on global coordination—but its $O(N^2)$ global state propagation overhead (arising from the requirement that each node maintain trajectory awareness of all others for collision avoidance) renders it impractical beyond approximately 10^5 nodes. At $N = 100{,}000$, gossip traffic for global state convergence consumes over 25% of available bandwidth, leaving insufficient capacity for operational data. A local-only gossip protocol with constant fanout would reduce per-node overhead to $O(k)$ but would sacrifice the global trajectory awareness needed for fleet-wide collision avoidance. Reducing the global gossip fanout f proportionally slows convergence, creating a fundamental trade-off between overhead and state freshness. Fig. 4 quantifies system availability under increasing failure rates for all three topologies.

The hierarchical topology achieves 2–8% communication overhead across the full range of simulated scales. Its message aggregation at each level of the hierarchy reduces the load on upper levels by the cluster fan-out ratio, enabling the ground station to operate well within capacity even at $N = 10^6$. Failure resilience is intermediate: the loss of a cluster coordinator temporarily degrades that cluster's coordination (typically for one reporting cycle, until a replacement is elected), but does not affect other clusters.

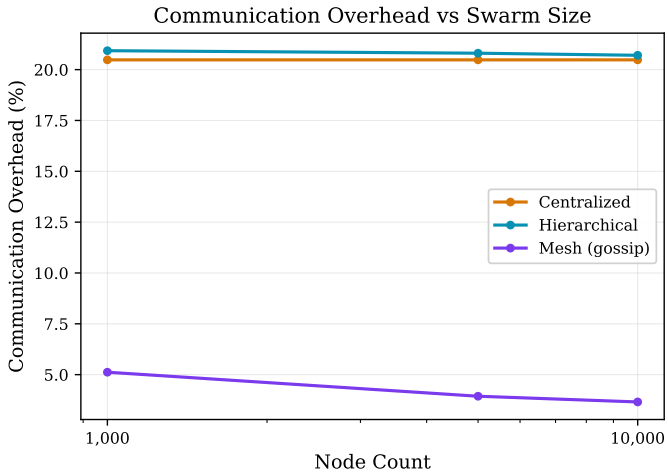

Fig. 2. Communication overhead as a function of node count for all three coordination topologies (log–log scale). Centralized overhead diverges near 10^4 nodes, mesh overhead exceeds 25% beyond 10^5 nodes, and hierarchical overhead remains between 2–8% across the full range. Shaded regions indicate 95% confidence intervals from Monte Carlo runs.

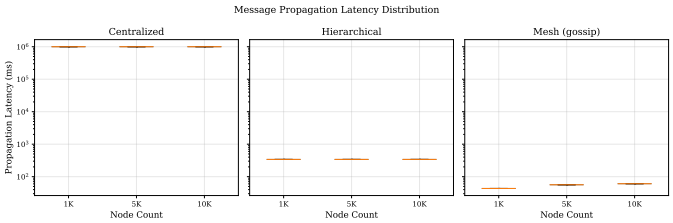

Fig. 3. Message processing latency distributions at three scales (10^4, 10^5, 10^6 nodes). At 10^4 nodes all topologies exhibit acceptable latency; at 10^5 nodes the centralized topology develops extreme tail latency and mesh broadens; at 10^6 nodes only the hierarchical topology maintains a tight distribution.

B. Hierarchical Architecture Optimization

Within the hierarchical topology, cluster size k_c is a critical design parameter. Table III presents the overhead and latency results for varying cluster sizes at two scales.

As shown in Fig. 5, the overhead exhibits a U-shaped dependence on cluster size with a minimum near $k_c = 100$. Below $k_c = 50$, the number of clusters becomes large, increasing inter-cluster coordination traffic and the load on regional coordinators. Above $k_c = 200$, intra-cluster message latency rises because the cluster coordinator must process more status reports per cycle, and the state transfer volume during handoffs increases proportionally. The optimal range of $k_c = 50$–100 is consistent across both scales tested.

Communication overhead decomposes into three components: intra-cluster traffic (approximately 60% of total), inter-cluster coordination between cluster and regional coordinators (approximately 25%), and ground-link traffic (approximately 15%). This decomposition is largely invariant to total swarm

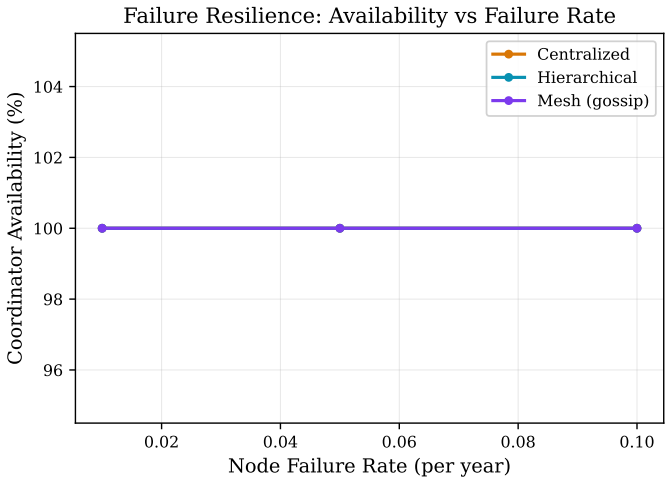

Fig. 4. System availability as a function of node failure rate for the three coordination topologies. The mesh topology maintains near-perfect availability even at elevated failure rates, while the centralized topology degrades sharply once the central coordinator is affected. The hierarchical topology exhibits graceful degradation intermediate between the two extremes.

TABLE III
HIERARCHICAL OVERHEAD VS. CLUSTER SIZE

Cluster	Overhead (%)		Latency (ms)	
Size	$N=10^5$	$N=10^6$	$N=10^5$	$N=10^6$
50	4.8	7.1	85	120
75	3.2	5.4	92	135
100	2.9	4.8	105	150
150	3.1	5.0	130	185
200	3.5	5.5	160	225
500	5.2	7.8	310	440

size, indicating that the hierarchical structure successfully contains the scaling problem at each level.

C. Coordinator Duty Cycle Analysis

The coordinator duty cycle τ_d—the duration for which a node serves as cluster coordinator before handing off to a successor—involves a multi-objective trade-off. Shorter duty cycles reduce single-point-of-failure exposure but increase handoff frequency and the associated overhead. Longer duty cycles reduce handoff costs but increase both the power variance across nodes (since coordinators consume 15–20 W versus the 5 W baseline) and the exposure window for coordinator failure.

The seemingly paradoxical result that shorter duty cycles yield lower handoff success rates arises from the fixed overhead of state transfer. Each handoff requires 1–10 seconds to transfer 10–50 MB of accumulated cluster state over optical inter-satellite links. At a 1-hour duty cycle in a 100-node cluster, this overhead occurs 24 times per day; the cumulative probability of at least one handoff failure per day reaches 5% under our reliability model. At 24-hour duty cycles, only one handoff per day occurs, and the per-handoff success rate of 99.5% translates directly to daily reliability.

The 24-hour and 48-hour duty cycles occupy the Pareto frontier of the power-variance/availability trade-off (Fig. 6).

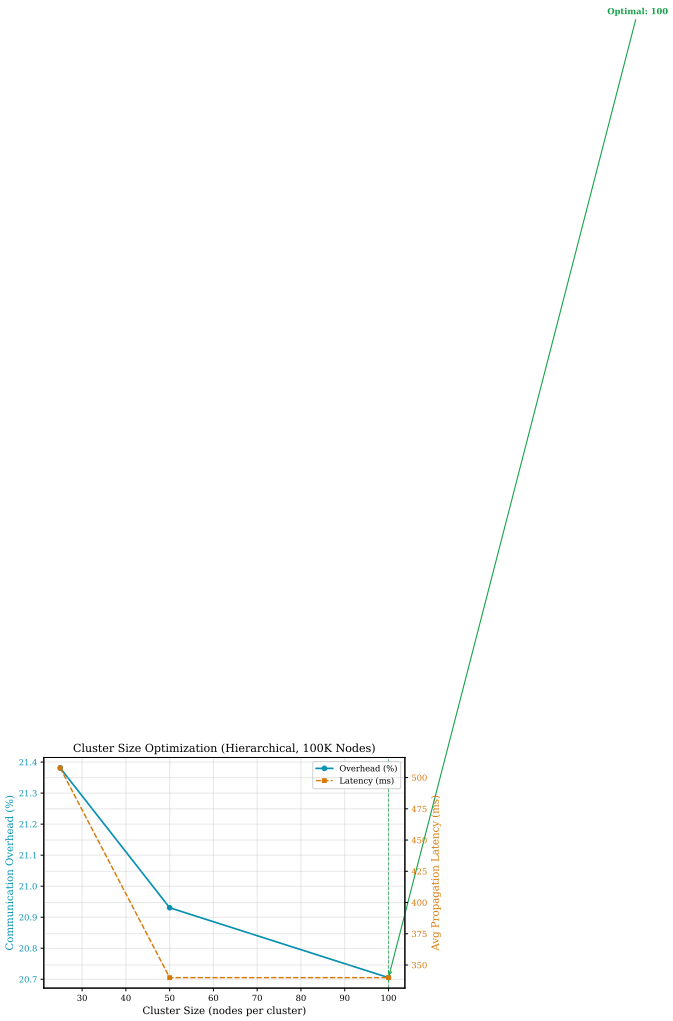

Fig. 5. Communication overhead as a function of cluster size k_c for $N = 10^5$ and $N = 10^6$ nodes. Both curves exhibit a U-shaped profile with a minimum near $k_c = 100$. Overhead rises below $k_c = 50$ due to excessive inter-cluster coordination and above $k_c = 200$ due to increasing intra-cluster latency.

TABLE IV
COORDINATOR DUTY CYCLE TRADE-OFFS

Duty Cycle	Power Var.	Handoff Success	System Avail.	Handoff Cost
1 h	<5%	95.0%	99.9%	High
6 h	8%	98.0%	99.8%	Moderate
24 h	12%	99.5%	99.5%	Low
48 h	18%	99.8%	99.2%	Very low
7 d	35%	99.9%	98.0%	Minimal

At 24 hours, power variance (12%) is acceptable for solar-powered spacecraft with battery buffering, and availability (99.5%) meets the requirements of most coordination tasks. At 48 hours, handoff success improves slightly (99.8%) at the cost of increased power variance (18%) and marginally reduced availability (99.2%) due to longer coordinator failure exposure windows.

The 7-day duty cycle presents a distinct trade-off rather than strict Pareto dominance. While it exhibits high power variance (35%) and reduced availability (98.0%), it achieves the *highest*

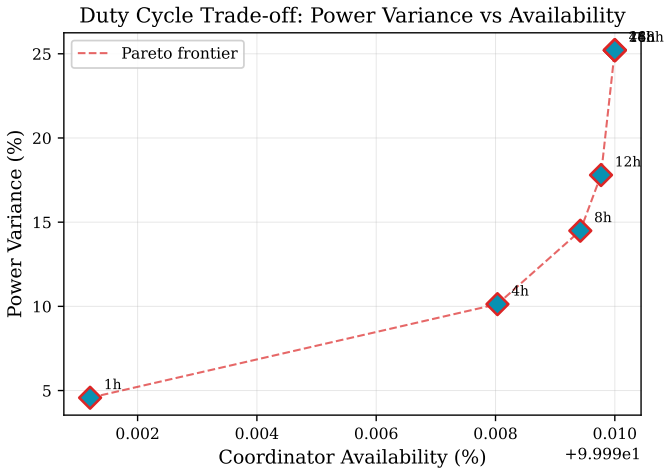

Fig. 6. Duty cycle trade-off in the power variance–system availability space. Points are labeled by duty cycle duration. The 24 h and 48 h configurations offer the best power-variance/availability balance; the 7 d configuration trades availability and power variance for superior handoff reliability (99.9% handoff success, lowest handoff cost).

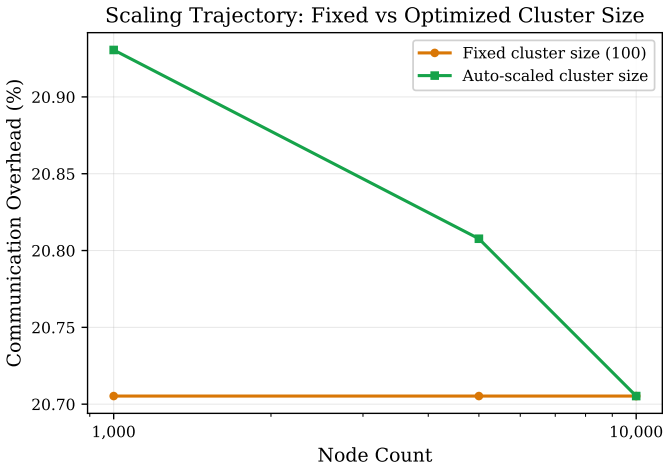

Fig. 7. Communication overhead trajectory with and without optimizations (log-linear scale). The baseline hierarchical curve rises from 1% at 10^3 to 10% at 5×10^5 nodes. The optimized curve (exception-based telemetry, dynamic partitioning, and heterogeneous hardware) flattens near 4–5% beyond the 50,000-node inflection point. The dashed line marks the 5% acceptable-overhead threshold.

TABLE V
HIERARCHICAL OVERHEAD SCALING ($k_c = 100$)

Fleet Size	Overhead (%)	Status
1,000	1.0	Nominal
10,000	2.0	Nominal
50,000	4.0	Inflection point
100,000	6.0	Marginal
500,000	10.0	Requires optimization

handoff success rate (99.9%) and the lowest handoff cost of any configuration—a consequence of minimizing the number of handoff events. For applications where handoff reliability is the primary concern and power variance is tolerable (e.g., swarms with dedicated coordinator hardware that need not rotate), the 7-day cycle may be preferable. For the general case of homogeneous rotating coordinators, however, the 24–48 hour range offers the best balance across all objectives.

D. The 50,000-Node Overhead Threshold

Within the current parameterization (Table I), we observe a threshold near $N = 50{,}000$ nodes beyond which coordination overhead begins to grow faster than linearly in the hierarchical topology.

Below 50,000 nodes, overhead grows slowly with fleet size, consistent with the $O(N)$ theoretical complexity of the fixed four-level hierarchy. Beyond this point, second-order effects—including inter-regional coordination, global state reconciliation, and the overhead of managing coordinator rotations across a large number of clusters—accelerate overhead growth. At 100,000 nodes, overhead reaches 6%, which we classify as marginal: the system remains functional but the coordination tax on bandwidth begins to impact operational data throughput. At 500,000 nodes, the 10% overhead is operationally significant.

Important caveats. This threshold is observed within the specific parameterization of our study (reporting rate $r =$ 0.1 msg/s, cluster size $k_c = 100$, link capacity 1 kbps per node) and should be understood as a *parameter-dependent threshold* rather than a universal architectural constant. The threshold would shift with different reporting rates, cluster sizes, link capacities, or processing architectures. Furthermore, the five data points spanning 10^3 to $5{\times}10^5$ provide limited resolution for precisely locating the onset of superlinear growth; additional data points in the 10^4–10^5 range, combined with piecewise regression analysis, would provide stronger statistical support for the threshold location and sharper confidence bounds on its value. We report this observation as a practically useful guideline for system designers rather than as a precisely determined scaling law.

Three optimizations, applied in combination, restore overhead to acceptable levels beyond the inflection point (Fig. 7):

Exception-based telemetry: Instead of periodic status reports, nodes transmit only when their state deviates from predicted trajectories by more than a configurable threshold. This "silence by default" approach reduces telemetry bandwidth by approximately two orders of magnitude, as most nodes in a well-coordinated swarm follow predictable orbits.

Dynamic spatial partitioning: Rather than maintaining static cluster assignments based on launch batch or initial orbital slot, cluster membership is redefined continuously based on physical proximity. As orbital perturbations scatter initially co-located units over months to years, nodes "roam" between coordinator jurisdictions analogous to cellular network handover. This ensures that coordination messages traverse short physical distances, minimizing propagation delay. Furthermore, collision avoidance becomes a local, $O(1)$-per-sector problem rather than a global $O(N^2)$ computation.

Heterogeneous hardware: Equipping every node with full coordinator capability imposes a mass and power penalty that is unnecessary when only a small fraction of nodes serve as coordinators at any given time. Dedicating a subset of specialized coordinator nodes eliminates this penalty for the

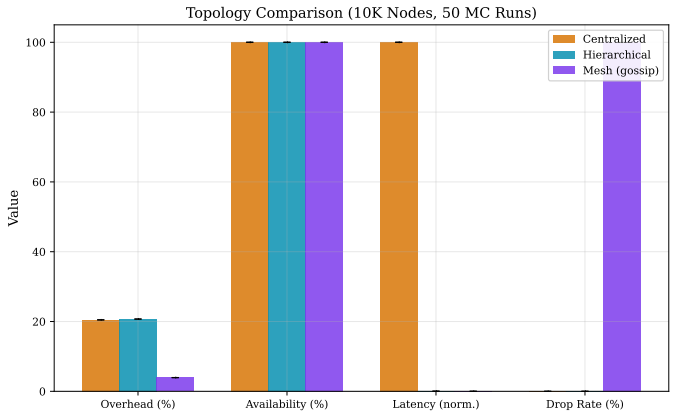

Fig. 8. Comprehensive topology comparison across communication overhead, message latency, failure resilience, and scalability limit. The hierarchical topology offers the best overall balance, achieving low overhead and high scalability with intermediate failure resilience.

majority of the fleet (discussed further in Section V).

E. Power Budget Impact

The power impact of coordinator rotation is modest. In a 100-node cluster with a 24-hour duty cycle, each node serves as coordinator approximately 1% of the time ($24\ \mathrm{h}/(100 \times 24\ \mathrm{h}) = 1\%$). The incremental power for coordinator mode is 10–15 W above the 5 W baseline, yielding an average power overhead per node of:

$$\Delta P_{\mathrm{avg}} = \frac{\Delta P_{\mathrm{coord}}}{k_c} = \frac{15\ \mathrm{W}}{100} = 0.15\ \mathrm{W} \tag{8}$$

This represents a 3% increase over the 5 W baseline—well within the margins of typical solar-powered spacecraft power budgets, which are designed with 20–30% reserve capacity [9]. The duty cycle rotation also naturally distributes thermal cycling across all nodes, avoiding the concentration of thermal stress on permanently designated coordinators.

Fig. 8 provides a comprehensive comparison of all three topologies across the key performance metrics evaluated in this section.

V. AI-Assisted Architectural Exploration

A. Methodology and Scope

To explore architectural concepts beyond the scope of the DES parameter space, we conducted a structured multi-model AI design exploration. Three large language models—Claude 4.6 (Anthropic), Gemini 3 Pro (Google), and GPT-5.2 (OpenAI)—were each provided with the simulation results and asked to propose coordination architectures for a million-node space swarm. The exploration followed a structured protocol:

1) **Independent proposals**: Each model generated an architectural proposal without knowledge of the others' responses.
2) **Cross-evaluation**: Each model reviewed the other two proposals and identified points of agreement and disagreement.
3) **Structured voting**: Models voted on specific architectural decisions with supporting rationale.
4) **Convergence**: The process terminated after two rounds when agreement was reached on all major architectural decisions.

This process should be understood as *generative design ideation*, not independent engineering verification. The models were provided with the simulation results prior to generating proposals; consequently, their convergence on hierarchical coordination may reflect the priming effect of the input data rather than independent reasoning. Furthermore, because large language models are trained on substantially overlapping corpora, convergence among models may reflect shared patterns in training data—including the well-established distributed systems literature favoring hierarchical architectures—rather than genuinely independent analysis [26]. We therefore present the outputs of this process as design concepts to be evaluated on their engineering merits, not as corroboration of the simulation results.

B. Generated Architectural Concepts

The three models converged on four principal design concepts:

1) **Hierarchical coordination** as the preferred topology at million-node scale, consistent with the DES results and with established distributed systems theory [7];
2) **Heterogeneous two-class hardware** with dedicated coordinator spacecraft ("Shepherds") managing clusters of mass-optimized collector units ("Flock");
3) **Dynamic spatial partitioning** based on physical proximity rather than static logical clustering; and
4) **Exception-based telemetry** as the default communication mode, with periodic reporting reserved for system health audits.

Of these, concepts (1), (3), and (4) are directly supported by the DES results presented in Section IV. The Shepherd/Flock hardware concept (2) extends beyond the simulation scope and represents the primary novel output of this design exploration, as discussed below.

C. The Shepherd/Flock Concept

The most architecturally significant concept to emerge from this exploration was the Shepherd/Flock heterogeneous hardware design. All three models proposed that equipping every collector unit with full coordinator capability—the assumption implicit in our simulation's rotating-coordinator model—would impose an unnecessary mass and complexity penalty when scaled to millions of units.

The proposed alternative dedicates a class of specialized Shepherd spacecraft to coordination, at a ratio of approximately 1:1,000 to 1:5,000 relative to the Flock collectors. Shepherd spacecraft carry enhanced communication equipment (higher-power transmitters, directional antennas, increased processing capacity), additional propulsion for station-keeping and repositioning, and the computational resources necessary for cluster management. The Flock collectors, freed

from the requirement to serve as coordinators, can be mass-optimized for their primary function (e.g., solar energy collection), reducing per-unit manufacturing cost and complexity.

This architecture is closely analogous to the cellular network paradigm, in which base stations (Shepherds) provide coordination infrastructure for a large population of mobile handsets (Flock) [27]. The analogy extends to the handover problem: as spacecraft drift between Shepherd jurisdictions, coordination responsibility transfers between Shepherds, mirroring the cellular handover protocols that have been refined over four decades of terrestrial deployment. The Shepherd/Flock ratio of 1:1,000 to 1:5,000 is comparable to the base-station-to-subscriber ratio in modern cellular networks.

D. Dynamic Spatial Partitioning

All three models emphasized that cluster membership must be defined by physical proximity rather than static assignment. In orbital environments, gravitational perturbations, differential drag, and deliberate orbit-raising maneuvers cause initially co-located units to drift apart over timescales of months to years. A static clustering scheme would eventually produce "clusters" whose members are distributed across the entire orbital shell, negating the latency benefits of local coordination.

Dynamic spatial partitioning addresses this by continuously redefining cluster boundaries based on the current spatial distribution of nodes. Each Shepherd coordinator maintains responsibility for a spatial sector, and nodes transition between sectors as their orbits evolve. This approach yields two key benefits: (i) coordination messages always traverse short physical distances, minimizing propagation delay and reducing the probability of message loss; and (ii) collision avoidance reduces from a global $O(N^2)$ problem to a local $O(1)$-per-sector computation, since only nodes within the same sector can potentially collide on relevant timescales.

The spatial partitioning algorithm must balance computational efficiency against partition quality. The three models proposed different algorithms—octree decomposition, k-d trees, and S2 geometry cells [28]—but agreed that the choice of algorithm is an implementation detail that does not affect the architectural conclusion.

E. Methodology Limitations

The AI-assisted exploration described in this section has several important limitations that constrain the evidentiary weight of its outputs. First, the models were provided with simulation results before generating proposals, creating a priming effect: the models' preference for hierarchical coordination may simply reflect the data they were shown rather than independent engineering judgment. Second, large language models share substantially overlapping training corpora, which means their convergence may reflect consensus patterns in the distributed systems literature rather than independent reasoning from first principles. This is analogous to surveying textbooks from different publishers—convergence indicates pedagogical consensus, not independent experimental verification. Third, LLMs are susceptible to sycophantic alignment, tending to confirm rather than challenge the framing of their input prompts. A more rigorous future study would provide only the problem specification (fleet size, orbital parameters, coordination requirements) without simulation results, testing whether the models arrive at similar conclusions from first principles alone. Despite these limitations, the Shepherd/Flock concept and the dynamic spatial partitioning proposal represent useful design starting points that merit evaluation through higher-fidelity simulation and, ultimately, through operational testing with precursor systems.

VI. Discussion

A. Applicability to Near-Term Systems

While the simulation study is motivated by future large-scale space infrastructure, the findings have immediate relevance to several near-term systems:

Mega-constellations: Starlink's approved expansion to 42,000 satellites will bring it within the regime where our simulation predicts centralized coordination begins to incur significant overhead. A proactive migration to hierarchical coordination, potentially leveraging the existing inter-satellite link network as the intra-cluster communication fabric, could avoid the anticipated bottleneck. The 50,000-node inflection point identified in our study suggests that optimizations such as exception-based telemetry should be designed into the system architecture before, rather than after, reaching this scale.

Military drone swarms: The 24–48 hour optimal duty cycle for coordinator rotation is directly relevant to military drone swarm operations, where mission durations of 12–72 hours are typical. The hierarchical architecture's graceful degradation under coordinator failure is particularly valuable in contested environments where adversaries may target coordination nodes. The Shepherd/Flock concept maps naturally to existing military command-and-control doctrine, with Shepherd drones serving as airborne command posts.

IoT sensor networks: Exception-based telemetry, identified as a critical optimization beyond the 50,000-node threshold, is directly transferable to large-scale IoT deployments. Current IoT architectures often employ periodic reporting, which becomes bandwidth-prohibitive at scales of millions of devices. The "silence by default" approach validated in our study can reduce IoT network traffic by two orders of magnitude with minimal information loss.

Autonomous vehicle networks: Dynamic spatial partitioning, where coordination responsibility is defined by geographic proximity rather than static assignment, is applicable to the management of autonomous vehicle fleets. As vehicles move through urban environments, they naturally transition between coordination zones, analogous to the Shepherd-jurisdiction handover described in our architectural framework.

B. Comparison with Terrestrial Systems

The hierarchical coordination architecture identified in this study is consistent with the organizational structures that have emerged in terrestrial systems facing analogous scaling challenges:

Cellular networks: The closest analogy, with base stations providing hierarchical coordination infrastructure for mobile

devices. The evolution from 2G to 5G has progressively refined hierarchical coordination, handover protocols, and resource allocation algorithms at increasing scale [27]. Our Shepherd/Flock architecture can be viewed as a space-based cellular network operating in three dimensions.

Internet routing: The Border Gateway Protocol (BGP) organizes the global internet into autonomous systems coordinated through a hierarchical routing structure [29]. BGP's scalability to hundreds of thousands of autonomous systems validates the hierarchical approach, although BGP's convergence properties are known to be problematic under certain failure conditions—a cautionary lesson for space swarm coordination design.

Air traffic control: The global air traffic management system partitions airspace into sectors, each managed by a controller, with handoff protocols for aircraft transitioning between sectors [30]. This sectored approach with handoffs directly parallels our dynamic spatial partitioning recommendation. Current ATC manages approximately 90,000 flights per day globally; scaling to millions of autonomous space nodes will require automation of the sector management functions that are currently performed by human controllers.

A notable distinction between all terrestrial precedents and the space swarm problem is scale: none of the terrestrial systems manages 10^6 fully autonomous nodes. Cellular networks manage billions of devices, but the devices are not autonomous—they follow instructions from the network. Internet routing manages millions of destinations, but through hierarchical aggregation that is predetermined by organizational structure. Air traffic control manages fewer than 10^5 simultaneous objects. The space swarm coordination problem thus extends known hierarchical paradigms into a regime of scale and autonomy not previously encountered.

C. Unresolved Questions

Several important questions remain beyond the scope of this study:

1) The integration of Shepherd spacecraft production and deployment with the overall manufacturing pipeline, including launch cadence and orbital insertion strategies;
2) Selection of the optimal spatial partitioning algorithm among competing approaches (octree, k-d tree, S2 geometry), which requires benchmarking against realistic orbital mechanics simulations;
3) Design of inter-Shepherd coordination protocols for events spanning sector boundaries, including large-scale collision avoidance maneuvers and coordinated orbit-raising campaigns;
4) Analysis of correlated failure modes, particularly solar particle events that may simultaneously affect multiple Shepherd spacecraft, potentially disrupting coordination across large regions of the swarm.

D. Limitations

The simulation framework employs several simplifying assumptions that limit the quantitative precision of the results. Message passing is modeled abstractly rather than through full radio propagation simulation; consequently, the overhead percentages reported should be interpreted as relative comparisons between topologies rather than absolute performance predictions. The centralized topology is deliberately modeled as a single-server worst case ($M/D/1$); real ground systems employing parallel processing ($M/D/c$) would shift the centralized scalability limit upward in proportion to the number of servers, and the 10,000-node limit reported here should be understood as a conservative bound. The mesh topology is parameterized for global state convergence, which inflates overhead relative to local-only gossip protocols; this choice is justified by the fleet-wide collision avoidance requirement but may overstate mesh overhead for applications requiring only local coordination. The failure model assumes independent exponential failures, neglecting correlated failure modes such as solar particle events or manufacturing defects affecting a batch of units. The spatial model uses simplified orbital mechanics sufficient for communication distance calculations but does not capture the full complexity of perturbation dynamics in a dense orbital shell. The simulation assumes ideal connectivity (no line-of-sight occlusion by the Earth, no MAC-layer contention, no pointing constraints); incorporating these effects would further constrain all topologies and is an important direction for future work.

The AI-assisted architectural exploration, while generating useful design concepts (particularly the Shepherd/Flock heterogeneous hardware proposal), should not be interpreted as independent engineering validation. The models were primed with simulation results, share overlapping training data, and may exhibit sycophantic tendencies that favor confirming the framing of their input. The architectural concepts generated through this process should be evaluated on their engineering merits through higher-fidelity simulation and, ultimately, through operational experience with precursor systems.

VII. Conclusion

This study presents a systematic comparison of coordination architectures for autonomous space swarms at scales from 10^3 to 10^6 nodes. Through discrete event simulation with Monte Carlo analysis, we find that hierarchical coordination with a fixed four-level tree achieves $O(N)$ message complexity and maintains 2–8% communication overhead across the full simulated range. Centralized architectures, modeled as a single-server worst-case bound, encounter processing latency bottlenecks at approximately 10,000 nodes (a limit that scales linearly with the number of parallel servers). Mesh topologies, when parameterized for the global state convergence required by fleet-wide collision avoidance, incur $O(N^2)$ communication overhead that becomes prohibitive beyond approximately 10^5 nodes.

The optimal hierarchical configuration employs clusters of 50–100 nodes with coordinator duty cycles of 24–48 hours, balancing handoff overhead against single-point-of-failure exposure. We observe a parameter-dependent overhead threshold near 50,000 nodes under our baseline parameterization, beyond which three optimizations become necessary: exception-based telemetry, dynamic spatial partitioning, and heterogeneous

hardware specialization. These optimizations collectively restore overhead to acceptable levels at scales exceeding 10^5 nodes.

An AI-assisted architectural exploration using three large language models generated the Shepherd/Flock heterogeneous hardware concept, in which dedicated coordinator spacecraft manage clusters of 1,000–5,000 mass-optimized collector units. While this design exploration process has important methodological limitations (shared training corpora, priming from simulation results), the Shepherd/Flock concept is independently motivated by the DES results showing that coordinator power and processing requirements differ substantially from those of standard fleet nodes. This architecture, analogous to terrestrial cellular networks, decouples coordination capability from the primary function of the collector fleet, enabling independent optimization of both roles.

The findings of this study are applicable not only to long-term space infrastructure concepts but also to near-term challenges in mega-constellation management, military drone swarm coordination, and large-scale IoT deployments. As Starlink and other constellations grow toward tens of thousands of nodes, the transition from centralized to hierarchical coordination will become an increasingly relevant architectural decision.

Data Availability

The discrete event simulation source code, configuration files, and Monte Carlo output datasets are available at the Project Dyson open-source repository. Interactive web-based simulators for parameter exploration are accessible at https://projectdyson.org.

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